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Effects of porous media thickness and its hydraulic gradient history on the formation of sand boils: Experimental investigation

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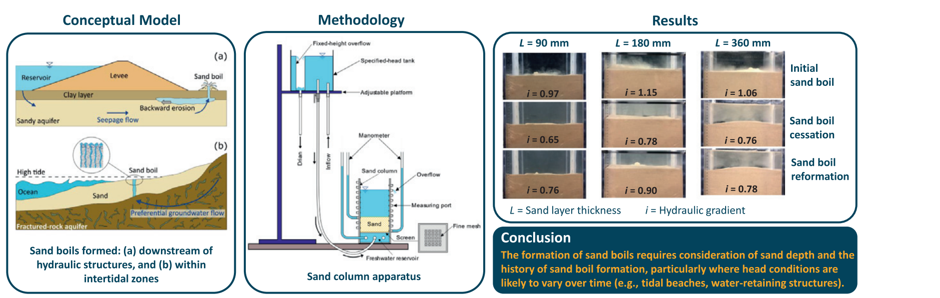
HIGHLIGHTS

- Laboratory experiments used to examine sand-depth and head-change effects on sand boil behaviour.
- Terzaghi's theory underestimates the critical hydraulic gradient (for sand boil formation) for thicker sand layers.
- The critical hydraulic gradient is lower for sand boil re-emergence under fluctuating head conditions.

GRAPHICAL ABSTRACT



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ABSTRACT

Sand boils occur where groundwater discharges to the land surface under sufficient hydraulic gradient to cause internal erosion and the upward transport of particles. A proper understanding of sand boil processes is essential in evaluating a wide range of geomechanical and sediment transport situations under which groundwater seepage occurs, such as the effects of groundwater discharge on beach stability. Although various empirical methods have been developed to estimate the critical hydraulic gradient (i_{cr}) leading to sand liquefaction, a prerequisite for sand boil occurrence, the effect of sand layer thickness and the implications of driving head fluctuations on the formation and reformation of sand boils have not been explored previously. This paper uses laboratory experiments to study sand boil formation and reformation for various sand thicknesses and hydraulic gradients to fill this knowledge gap. Sand layer thicknesses of 90 mm, 180 mm and 360 mm were adopted in evaluating sand boil reactivation, which was created by imposing hydraulic head fluctuations. While the first experiment (i.e., 90 mm sand layer) yielded a value for i_{cr} smaller (by 5%) than Terzaghi's (1922) value, the same theory underestimated i_{cr} by 12% and 4% for 180 mm and 360 mm sand layers, respectively. Moreover, i_{cr} needed for the reformation of sand boils decreased by 22%, 22% and 26% (relative to i_{cr} applicable to the initial sand boil) for the 90 mm, 180 mm and 360 mm sand layer thicknesses, respectively. We conclude that the formation of sand boils requires consideration of sand depth and the history of sand boil formation, particularly in relation to sand boils that form (and potentially reform) under oscillating pressures (e.g., tidal beaches).

1. Introduction

Groundwater seepage through porous media may lead to the dislodgement and upward movement of particles, causing subsurface erosion (Ojha et al., 2003). Where hydraulic gradients and the upward seepage of groundwater in granular sediments are sufficiently intense, "sand boils" may occur (Werner et al., 2020). Fig. 1 shows two common types of sand

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Notation

ε [−] pipe wall roughness
μ [LT^{-2}] dynamic viscosity of water
ρ_s [ML^{-3}] soil density
ρ_w [ML^{-3}] density of the water
ψ_b; ψ_t [L] pressure head at the bottom and top of the sand layer, respectively
Δh [L] head difference between the bottom and top of the sand layer
A [L^2] cross-section area perpendicular to the flow
C_u [−] uniformity coefficient ($= d_{60}/d_{10}$)
d [L] particle size
$d_{c,15}$ [L] grain diameters for which 15% of grains by mass of the coarse fraction of the soil are smaller
$d_{f,85}$ [L] grain diameters for which 85% of grains by mass of the fine fraction of the soil are smaller
d_m [L] mean grain size calculated as $\log_2 d_m = (\log_2 d_{16} + \log_2 d_{50} + \log_2 d_{84})/3$
d_{10}; d_{16}; d_{50}, d_{60}; d_{84} [L] grain diameters at which 10%, 16%, 50%, 60%, and 84% (respectively) of a sample's mass is comprised of finer particles
F [−] mass fraction smaller than a particle size of d
G_s [−] soil specific gravity ($= \rho_s/\rho_w$)
h_b; h_t [L] heads at the bottom and top of the sand layer, respectively
H [−] incremental mass fraction measured between particle sizes d and $4d$
H_{min} [−] minimum of H
H/F [−] index to evaluate the soil's internal stability
$(H/F)_{min}$ [−] minimum of H/F
i [−] hydraulic gradient
i_{cr} [−] critical hydraulic gradient
L [L] sand layer thickness
n [−] soil total porosity
q [LT^{-1}] specific discharge or Darcy velocity ($= Q/A$)
Q [L^3T^{-1}] overflow discharge from the sand column
z_b [L] elevation of manometer ports on the front side of the sand tank
z_t [L] elevation of the base of the sand layer overlaying the screen

boils and the associated subsurface erosion, through which sand and water are expressed at the ground surface through single vents or clusters of vents.

A comprehensive understanding of the processes associated with sand boil formation and the accompanying rates of sediment and groundwater discharge is a critical first step in monitoring and managing hydraulic structure stability caused by the associated internal erosion, which is considered as the primary failure mode for dams and levees (Van Beek et al., 2011; Semmens and Zhou, 2019). The loss of sediment beneath or within hydraulic structures may lead to backward erosion (Fig. 1a), whereby a preferential discharge pathway may occur (and grow) in the form of a pipe, potentially causing accelerating sediment losses and "piping failure" of the structure if left unmitigated (Yang and Wang, 2017). Sand boils also occur in the polder regions of the Netherlands, where the groundwater expels sand (and saltwater) to the surface through fractures in overlying peat layers (De Louw et al., 2010), leading to volcano-shaped features at the boil vent. Where these occur in Dutch polders, intensive agriculture is threatened by the adverse influence of saltwater on soil productivity (e.g., De Louw et al., 2013). In those cases, the release of sand may exacerbate saltwater flow pathways to the land surface due to the undermining of surface peat layers. Sand boils have also been observed adjacent to river embankments, such as the Waal River (the Netherlands) and the Mississippi River (USA) (Robbins et al., 2020). Moosdorf and Oehler (2017) observed sand boils in the intertidal zone at Plage de Vaiva (Tahiti) and Ngobaran Beach (Indonesia). These were caused by submarine groundwater discharge to the shoreline. Sand boils are also evident within the intertidal zones of South Australia (e.g., Ramirez-Lagunas et al., 2023), where localized discharge from fractured and faulted aquifers creates high hydraulic gradients within overlying sand bodies (e.g., Fig. 1b). Coastal sand boils may potentially enhance beach erosion, although links between beach geomorphology and sand boils have not been studied.

Sand boil formation requires that liquefaction has occurred, which arises when the weight of an unconsolidated soil body is matched by the upward forces from flowing groundwater, leading to a loss of soil strength. Terzaghi (1922) was the first to propose a theorem for predicting the critical hydraulic gradient (i_{cr} [−]) causing liquefaction, where the effective stress (i.e., the mechanical stress between soil particles) is reduced to zero. Subsequently, various methods have been developed for predicting i_{cr}, which are commonly used to determine the occurrence of sand boils (e.g., Yang and Wang, 2017; Petrula et al., 2019). Yang and Wang (2017) offer a review of methods for calculating i_{cr}, and compared 44 measured values of i_{cr} and four different methods of predicting i_{cr} (i.e., Terzaghi, 1922; Wu, 1980; Liu, 1992; Zhou et al., 2010). They showed that none of the methods was applicable to all types of soils, concluding that Terzaghi's (1922) theory predicted the best results for i_{cr} when the soils were internally stable (i.e., soils in which small particles are unlikely to be mobilized by seepage flow or vibration; Kenney and Lau, 1985).

The vertical subsurface transport of soil particles within a soil mass, leading to sand boils, may occur due to several mechanisms that are likely to require hydraulic gradients (i [−]) that differ from those used for the prediction of sediment fluidization, as outlined in the review by Werner et al. (2020). They recognized two alternative conceptual models for particle movement, namely through the immobile pore network (whereby the soil body remains structurally competent), and via preferential pathways arising from fluidization. The former is referred to as "suffusion" which is more likely to occur in internally unstable soils, where particles smaller than the pore spaces are transported with groundwater flow (Fell and Fry, 2013; Fannin and Slangen, 2014). In contrast to suffusion, fluidization leads to localized discharge (i.e., "sand boils"; TACFD, 1999). Where this occurs as more widespread fluidization, rather than an individual vent, this is referred to as "boiling sand", in which the sediment surface appears as a boiling fluid (Kovacs, 1981; Werner et al., 2020). Werner et al. (2020) compared previous laboratory data with existing theory and showed that i values were required that were larger than the theoretical values of Terzaghi's (1922) i_{cr} to cause boiling sand within laboratory situations, including for both internally stable and unstable soils. However, published laboratory data confirmed that i values close to Terzaghi's (1922) i_{cr} are required for sand boil formation in internally stable soils (e.g., Yang and Wang, 2017; Petrula et al., 2019).

Various laboratory- and field-scale experiments have been conducted to investigate backward erosion, internal erosion, and sand boil formation. For example, Van Beek et al. (2011) undertook laboratory experiments to evaluate the effect of sand characteristics (e.g., grain size distribution, soil relative density and permeability) on the initiation and continuation of backward erosion processes leading to the failure of a levee. However, they could not find a clear relationship between sand characteristics and erosion patterns. For example, different erosion patterns were observed in their laboratory experiments in soils with low relative density to those with medium and high relative density. Robbins et al. (2020) conducted field measurements to ascertain the pressures and flow rates of sand boils that developed along the banks of the Mississippi and Waal Rivers. They compared measured head losses in the sand boils with their proposed theoretical model and showed that their developed theory could accurately predict the head loss distribution within the sand boils. Robbins et al. (2020) concluded that head losses (in the vertical direction) in sand boils are not constant, and rather, they depend on the throat diameter of the sand boil, along with the grain size and the fluid flow rate.

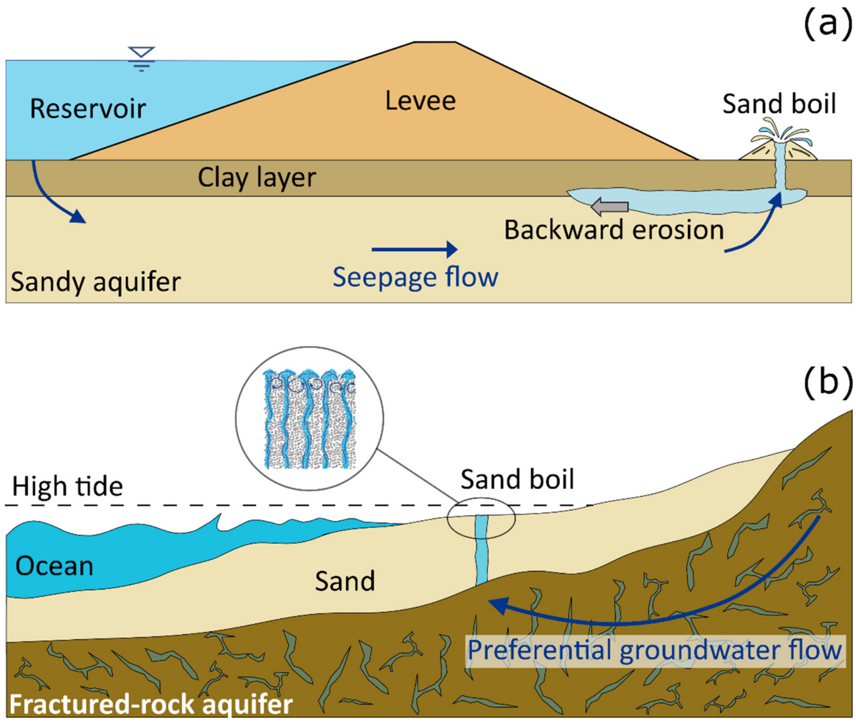

Fig. 1. Schematic illustration of sand boils formed: (a) downstream of hydraulic structures, and (b) within intertidal zones.

Laboratory experiments undertaken by Skempton and Brogan (1994) in sandy gravels showed that for internally stable soils, measured values of *i* equal to Terzaghi's (1922) i_{cr} were required for sand boils to occur. Wan and Fell (2008) conducted laboratory experiments to study the internal stability of silt-sand-gravel and clay-silt-sand-gravel soil mixtures subjected to various hydraulic gradients. Their soil configurations were gap-graded, with at least one range of particle sizes missing in the soil body. They found that suffusion occurred at a much lower *i* than Terzaghi's (1922) i_{cr} for many internally stable soils. Ahlinhan and Achmus (2010) conducted an experimental study to quantify the value of *i* required to cause suffusion in both internally stable and unstable soils, considering both upward and horizontal seepage flows. Their results showed that suffusion happened in internally stable soils at values of *i* that were slightly lower (~90%) of Terzaghi's (1922) i_{cr}. They also considered the effect of compaction on the soil samples and found that *i* required for suffusion for soils on the border between internally stable and unstable states is highly dependent on the soil relative density, whereby *i* required for suffusion in dense soils was twice that required for soils with medium density.

In both cohesive (e.g., silts and clays) and non-cohesive (i.e., sands and gravels) soils, heave (also known as *blow-out*) may occur if the upward hydraulic gradient is sufficiently steep. Heave refers to the rise of an intact soil body caused by seepage flow, and can theoretically occur at Terzaghi's (1922) i_{cr} (e.g., Werner et al., 2020). However, FEMA (2015) suggested to use of the term *heave* only for non-cohesive soils for consistency with Terzaghi's (1922) theory. The term *blow-out* was used by FEMA (2015) to describe the uplift of a low-permeability surface layer (confining or blanket layer) underlain by a more pervious layer, where the uplift is caused by high seepage pressure in the pervious layer. The same terminology suggested by FEMA (2015) is adopted in this study.

In cohesive soils (e.g., plastic clays), significantly higher *i* than Terzaghi's (1922) i_{cr} may be required to move clay particles due to electrochemical forces that hold particles together. Hence, FEMA (2015) suggested to use Terzaghi's (1922) equation to calculate i_{cr} only for non-cohesive soils. In calculating a safety factor against blow-out of a cohesive soil layer, FEMA (2015) offered alternative formulations involving either the effective stress or the total stress in the force balance analysis. In cohesive soils, blow-out occurs in places where the soil is weaker, leading to uplifting and/or cracking (i.e., hydraulic fracturing) that causes concentrated seepage flow in the form of pin boils (i.e., boils that discharge only water to the land surface) or sand boils (where both sand and water are expelled to the land surface), both of which were preceded by blow-out within low-lying areas of the Netherlands as described by De Louw et al. (2010, 2013).

Heave occurring in non-cohesive soils is usually accompanied by the subsequent development of sand boils (Wan and Fell, 2004). For example, Yang and Wang (2017) conducted a series of laboratory experiments on sand specimens subjected to upward seepage and observed heave before sand boil formation. Fleshman (2012) conducted laboratory experiments on sandy soils under upward flows and observed that sand boils were more commonly formed in well-graded soil samples (i.e., soil that contains a wide range of particle sizes) than in uniform soils (i.e., all particles in the soil are around the same size and shape). Fleshman (2012) observed values of *i* required for the occurrence of *initial* heave (i.e., the incipient motion of the soil particles on the sample surface and initial loosening of the topsoil), sand boil development and *total* heave (i.e., the entire soil sample moves upward) were higher than those predicted by Terzaghi's (1922) equation. Schmertmann (2015) argued that the higher *i* values for sand fluidization events observed in Fleshman's (2012) experiments were caused by sample side shear attributable to the obstruction to water and particle movement caused by the pore pressure probes, creating arching effects (i.e., pressure transfer between a moving mass of soil and adjoining stationary parts; Terzaghi, 1943). Hence, the results presented by Fleshman (2012) may be non-conservative (i.e., overestimation of i_{cr}) relative to the expected values of *i* applicable to the onset of the same phenomenon under field conditions. Petrula et al. (2019) also conducted experiments on upward seepage to study heave (fluidization). They used glass beads to maintain control over the grain size distribution. Comparison between their experimental results with various theoretical relationships (i.e., Terzaghi, 1922; Knorre, 1925; Zamarin, 1931) found that Terzaghi's (1922) equation provided the best agreement with *i* values obtained for the onset of heave.

Most experimental studies of sand boil formation explore the effects of the grain-size distribution, and therefore, it is presently unclear as to whether *i* required for sand boil formation differs when the sand body previously contained a sand boil (i.e., as might occur under fluctuating head gradients, such as in inter-tidal zones). This is notwithstanding that, intuitively, at least, a sediment body is likely less compacted in the vicinity of a site where a sand boil recently occurred. Moreover, none of the previous equations for estimating i_{cr} (see Table 1) consider the sand thickness or the antecedent conditions with respect to sand boil formation, even though Ahlinhan and Achmus (2010) showed that compaction, which can be influenced by the layer thickness, might be important for suffusion (typically a precursor to boil formation in laboratory experiments) (e.g., Wan and Fell, 2004; Ahlinhan and Achmus, 2010).

The current study focused on two key knowledge gaps in the current understanding of sand boil formation, as indicated by the following two objectives: (1) assess the effect of sand layer thickness on sand boil formation; and (2) examine the reformation of sand boils under variable driving head conditions. Sand boil formation will be explored under controlled conditions using laboratory experimentation. The emergence and re-emergence of sand boils under varying heads are likely to be critical for sand transport processes occurring in tidal beaches where sand boils have been observed (e.g., Ramirez-Lagunas et al., 2023). This is also likely to be important more generally for hydraulic structures where head conditions may vary in time.

2. Methodology

2.1. Theoretical methods

In a comparative study, Petrula et al. (2019) used 177 measured values of i_{cr} and compared these with the three analytical methods of predicting i_{cr} that are summarized in Table 1, in which n [−] is the soil's total porosity and G_s [−] represents the soil specific gravity obtained by dividing the soil density (ρ_s [ML^{-3}]) by the density of the water (ρ_w [ML^{-3}]).

2.2. Experimental apparatus and material

A schematic of the sand column apparatus utilized in this study to analyze sand boil formation is shown in Fig. 2. The sand column was constructed from 9 mm clear Perspex sheets, with a total internal height of 601 mm and an internal cross-section of 280 mm × 280 mm. A freshwater reservoir with a 90 mm internal height was created beneath the sand layer, which connected to the reservoir through a horizontal screen (Fig. 2) comprising a 9 mm Perspex sheet with twenty-five laser-cut holes, each 20 mm diameter, covered by a fine mesh. The freshwater reservoir at the bottom of the sand column ensured relatively uniform pressure across the sand layer base. Each column side had ten push-in connections ("Measuring ports" in Fig. 2) of 10 mm internal diameter and spaced evenly at 50 mm intervals. These allowed pressure head measurement using manometers or pressure sensors (devices were placed flush with the sand tank side to avoid the disturbances to flow reported by Schmertmann (2015)). The push-in connections were protected from sand clogging by a fine mesh attached to the inner sides of the sand column. Two manometers were connected to the freshwater reservoir via 20 mm access holes (Fig. 2) to measure the pressure head in the freshwater reservoir. The freshwater had a density ρ_w = 996 ± 1 kg/m^3 that was measured using a density hydrometer (Carlton Glass Co. Pty Ltd). Freshwater was supplied via a specified-head tank placed on an adjustable platform that allowed for smooth changes to the driving head during experiments. The specified-head tank received inflow from a 10 mm diameter inlet tube connected to the mains (i.e., via a tap). Outflow in the specified-head tank occurred through a fixed-height overflow, thereby maintaining the water level in the specified-head tank. Inflow to the sand column was supplied through a 10 mm tube connecting the specific-head tank to the freshwater reservoir at the base of the sand column (Fig. 2). Discharge from the sand column occurred via a pipe connected to one or two (depending on the discharge rate) of the measuring ports ("Overflow" in Fig. 2). Although the intent was to maintain a constant water level in the sand column above the sand surface during the experiments, some differences in the level of water above the sand (with different flow rates) were unavoidable, so this was monitored and considered in hydraulic calculations.

The sand (grade AIS70) used in the experiments was supplied by Adelaide Industrial Sands Pty Ltd. (South Australia). Grain size analysis indicated that the sand was fine (d_m = 0.18 mm) and uniformly graded (C_u = 1.64), with G_s assumed to be 2.65. Here, C_u (= d_{60}/d_{10}) is the uniformity coefficient [−], where a value 6 indicates a uniformly graded porous medium (Das, 2019). d_m [L] is the mean grain size calculated as $\log_2 d_m$ = ($\log_2 d_{16}$ + $\log_2 d_{50}$ + $\log_2 d_{84}$)/3 (Folk and Ward, 1957). d_{10}, d_{16}, d_{50}, d_{60} and d_{84} [L] are the grain diameters at which 10%, 16%, 50%, 60% and 84% (respectively) of the soil mass are finer. The values of d_{10}, d_{16}, d_{50}, d_{60} and d_{84} for the sand were obtained from the grain size distribution curve shown in Fig. 3a. The value of n of the experimental sand was 0.38 ± 0.01, determined by the water-saturation method (Fetter, 2001).

The internal stability of the sand used in the laboratory experiments of this study was assessed by three different criteria (i.e., Istomina, 1957; Kezdi, 1979; Kenney and Lau, 1986) that have been used in previous studies (e.g., Skempton and Brogan, 1994; Wan and Fell, 2008; Yang and Wang, 2017). Istomina (1957) classified soils based on the values of C_u, adopting three groups, namely, internally stable (C_u ≤ 10), transitional stability (10 ≤ C_u ≤ 20), and internally unstable (C_u ≥ 20). Kenney and Lau (1985, 1986) proposed the H/F [−] index to evaluate the soil's internal stability, where F [−] is the mass fraction smaller than a particle size of d [L], and H [−] is the incremental mass fraction measured between particle sizes d and $4d$ (Fig. 3a). According to Kenney and Lau (1986), the soil is considered internally stable if the minimum value of H/F (i.e., $(H/F)_{min}$ [−]) is ≥ 1. Kezdi (1979) proposed to split the grain size distribution of the soil into fine and coarse parts (using an arbitrary particle size to distinguish fine and coarse parts) to assess the stability of the soil using Terzaghi's filter rule (Terzaghi and Peck, 1948) of $d_{c,15}/d_{f,85}$ ≤ 4, where $d_{c,15}$ and $d_{f,85}$ [L] are the grain diameters for which 15% of grains by mass of the coarse fraction of the soil and 85% of the fine fraction of the soil are smaller, respectively. Chapuis (1992) stated that Kezdi's (1979) criterion means that the soil is internally stable if the minimum value of H (i.e., H_{min} [−]) is ≥ 15% for all values of F. Fig. 3b shows both Kenney and Lau's (1986) and Kezdi's (1979) soil stability criteria and corresponding values of F and H for the sand used in the experiments. The values of $(H/F)_{min}$ and H_{min} for the experimental sand were obtained as 2.1 and 68%, respectively. Hence, the sand used in this study is categorized as internally stable based on the criteria of both Kenney and Lau (1986) and Kezdi (1979) (Fig. 3b). The sand is also considered internally stable based on Istomina's (1957) criterion of C_u (= 1.68) < 10.

2.3. Experimental procedure

In order to achieve a relatively homogeneous medium, the sand layer of thickness L [L] was packed into the sand column using the wet-packing method, which has been widely employed in previous experimental studies (e.g., Werner et al., 2016; Jazayeri et al., 2021). In the wet-packing method, the sand was incrementally added to the water-filled sand column and gently mixed while settling to eliminate any air encapsulation and sand layering.

Experiments consisting of three main stages (i.e., Stages 1, 2 and 3) were conducted in the sand column to assess the effects of i variations on the formation and reformation of the sand boils. In Stage 1, i was incrementally increased by raising the adjustable platform height (i.e., the driving head). At each head, the experiment continued for around 10 min to ensure that the system was hydraulically stable, as determined by time-invariant water levels in manometers and the overflow discharge (Q [L^{3}T^{-1}]) from the

Table 1
Equations for the critical hydraulic gradient (i_{cr}).

Reference	Equation
Terzaghi (1922)	$i_{cr} = (1-n)(G_s-1)$
Knorre (1925)	$i_{cr} = G_s-1$
Zamarin (1931)	$i_{cr} = (1-n)(G_s-1)+0.5n$

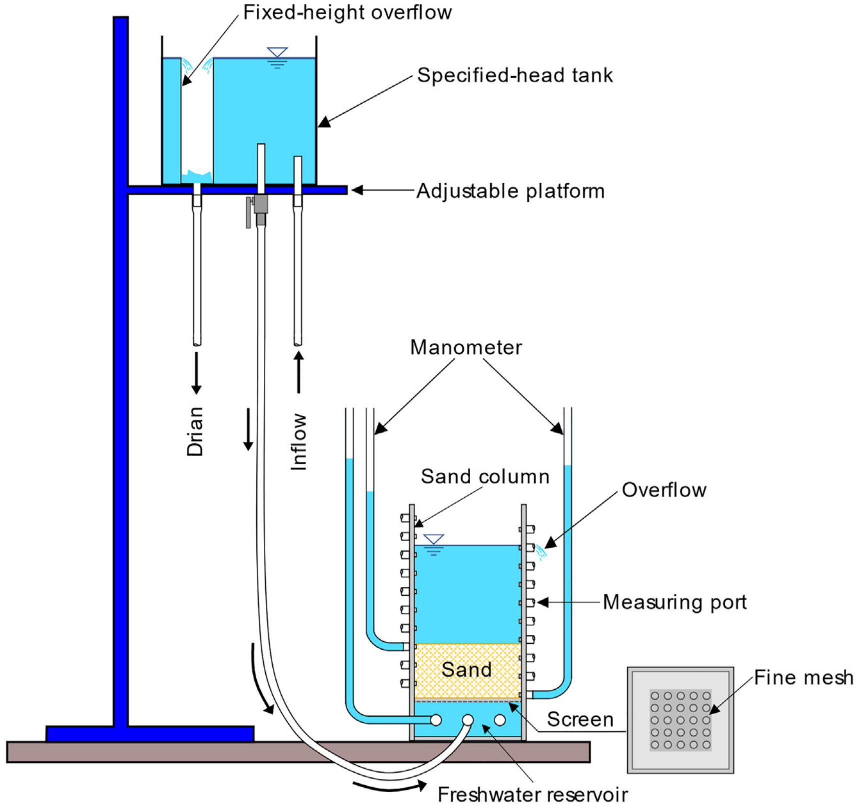

Fig. 2. Schematic diagram of the sand column apparatus setup.

sand column. When the water levels in the manometers stabilized, the pressure head at the bottom of the sand layer (ψ_b [L]) was measured using the manometers connected to the freshwater reservoir (the average value of the two manometers was adopted), while the pressure head at the top of the sand layer (ψ_t [L]) was obtained by measuring the water level in the sand column (with respect to the sand layer base). The elevations of manometer ports on the front side of the sand tank (the centroid of the access holes) and the base of the sand layer overlaying the screen (i.e., z_b and z_t [L], respectively) were also measured from a datum (i.e., $z = 0$ at the base of the sand column; Fig. 4) as 54 mm and 108 mm for z_b and z_t, respectively. Heads were obtained as the sum of pressure and elevation heads, where the heads at the bottom and top of the sand layer are given by h_b and h_t [L], respectively. Although a small head loss may have occurred due to the passage of inflowing water through the lower screen, it was presumed to be negligible (i.e., in the order of 10^{-3} mm based on the head loss through the screen's holes using the formulae of White, 2016). Additionally, any head losses incurred from the upward discharge of water within the reservoir above the sand sample were also neglected. Hence, we assumed that the head in the lower freshwater reservoir represented the head at the base of the sand layer (i.e., h_b), and the water level in the upper reservoir indicated the head at the top of the sand layer (i.e., h_t). The geometric variables of the sand boil experiments are shown in Fig. 4.

The vertical hydraulic gradient across the sand layer was obtained from:

$$i = \frac{\Delta h}{L} = \frac{h_b - h_t}{L} \tag{1}$$

where Δh [L] is the head difference between the sand layer bottom ($h_b = \psi_b + z_b$) and top ($h_t = \psi_t + z_t$).

Two cameras, one in front of the sand column and one above it, were used to monitor changing conditions during the experiments. Any observable changes were recorded and described (see Tables S1, S2 and S3 in the Supplementary material). The triggers for progression from one stage of the experiment to the next were as follows. Stage 1 continued until the

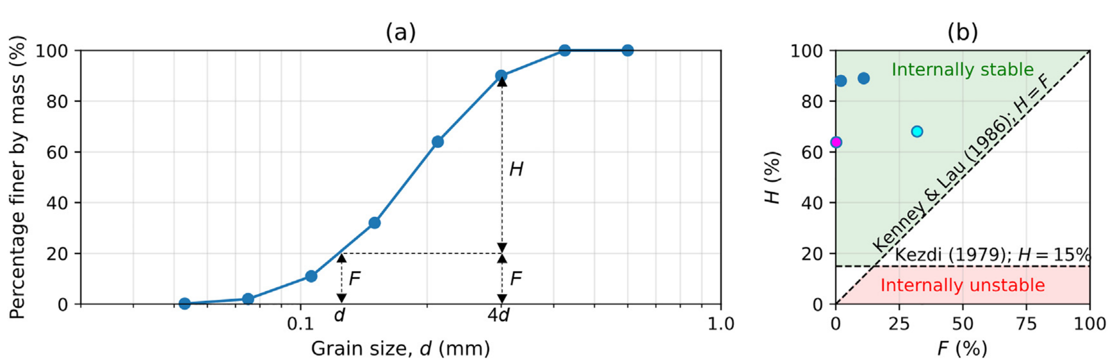

Fig. 3. Sand properties in terms of: (a) grain size distribution curve, and (b) the $F-H$ diagram to assess soil stability. In panel (b), the magenta and cyan circles indicate H_{min} (= 68%) and $(H/F)_{min}$ (= 2.1), respectively. The green- and pink-filled areas in panel (b) identify internally stable and unstable soil types (respectively) based on both Kenney and Lau's (1986) and Kezdi's (1979) soil stability criteria.

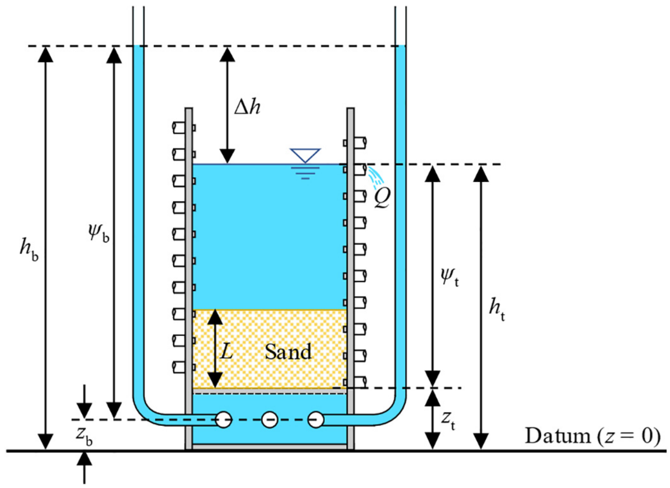

Fig. 4. Geometric variables of the sand boil experiments.

sand boil(s) appeared on the sand surface. After the formation of the sand boil(s) (and a short period thereafter of stabilization), i was incrementally decreased by lowering the height of the adjustable platform (Stage 2 of the experiments). Stage 2 proceeded until the sand boil (s) disappeared from the sand surface. Stage 3 (increasing i) commenced immediately after the sand boil(s) disappeared (i.e., at the end of Stage 2) and continued until the sand boil(s) reformed. After the reformation of the sand boil(s), Stage 3 continued until the adjustable platform reached its maximum height (i.e., ~2 m above the datum).

The three-stage experiments described above were applied to sand bodies of three different thicknesses, namely L = 90 mm (Experiment A), 180 mm (Experiment B) and 360 mm (Experiment C). To ensure that the mechanical properties of the sand were consistent between experiments, the same sand and packing technique were used for each experiment.

3. Results and discussion

The measured values of h_b, h_t, i, Q and any observed changes in the sand layer condition for Experiments A, B and C (i.e., L = 90 mm, 180 mm and 360 mm, respectively) are listed in Tables S1, S2 and S3 in the Supplementary Material. Fig. 5 presents the key observations from Experiment A, including the time series of i and corresponding Q, as well as the changes in the sand surface conditions at the end of Stages 1 and 2 and during Stage 3 when the sand boil reformed.

Aside from the transience in experimental measurements given in Fig. 5, a number of key qualitative observations were made from visual inspections of the experiments. For example, in Experiment A (i.e., L = 90 mm), no obvious changes in the sand layer condition were observed when i was increased up to 0.69 during Stage 1. The first visible sand particle movements (which can be interpreted as suffusion), in the form of very small craters on the sand surface and cloudiness in the water column, occurred during Stage 1 when i reached 0.76. The initial sand boil formed suddenly on the sand surface when i reached 0.97 at the end of Stage 1. No obvious heave (i.e., expansion in the sand layer thickness) was detected before the formation of the initial sand boil, while a significant loss of very fine material, which was apparent through the cloudiness of water above the sand layer, was observed (Fig. 5b). The location of the sand boil was mostly constant during the experiment, occurring in approximately the middle of the sand column cross-section. In Stage 2 (reducing i by lowering the driving head), smaller values of i (in the range of 0.67–0.73) than that required for the initial sand boil formation (i.e., i = 0.97) produced the sand boil, albeit with less vigorous flow and sand ejection. Dropping i to 0.65 caused sand boil cessation, marking the end of Stage 2. After the sand boil stopped, Stage 3 started, and i was increased again until the sand boil reformed, which occurred at i = 0.76. Stage 3 then continued until the end of the experiment.

Increases to the driving head during Stage 3 produced complex responses in i, which remained in the range of 0.65 to 0.76 (Fig. 5a). This is incongruent with Darcy's Law, but nevertheless expected given that boil discharge is expected to depart from classical porous media flow. Observations of complex responses in i to driving head changes can be attributed to three main factors: (1) widening of the boil aperture with increased driving

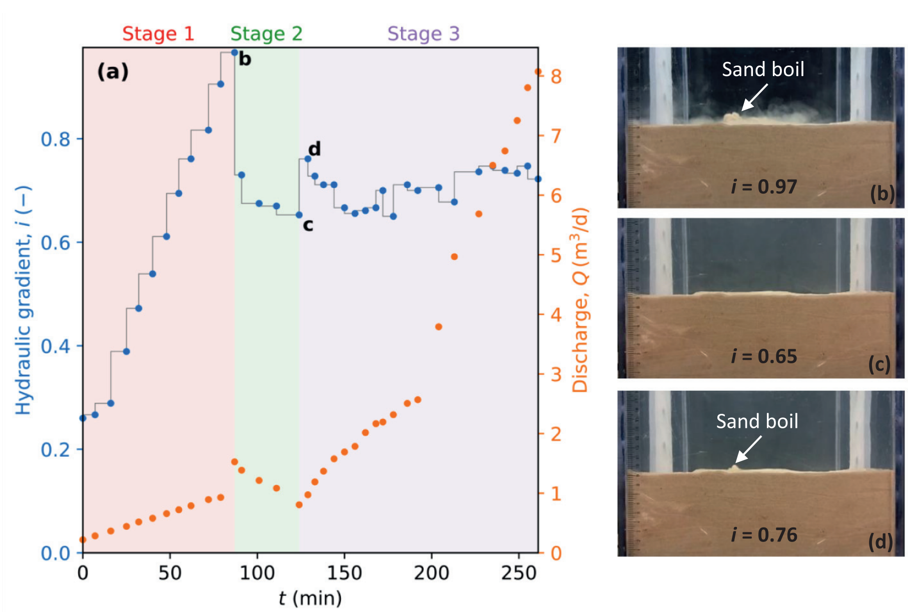

Fig. 5. Experiment A (L = 90 mm) results in terms of: (a) time series of i (blue circles) and corresponding Q (orange circles). Photographs of three key sand layer conditions at: (b) the first sand boil formation (the end of Stage 1), (c) sand boil cessation (the end of Stage 2), and (d) sand boil reformation during Stage 3. The light red, light green and light purple shaded areas in (a) represent Stages 1, 2 and 3 of the experiment, respectively.

head, thereby allowing a higher flux to pass through the sand mass for a given i. This serves to lower the resistance to flow, allowing a greater Q to pass through the sand for a given i (this is discussed in greater detail below); (2) greater head losses within the experimental pipework and apparatus more generally as the flow (and driving head) increased. Head losses were calculated (not shown here for brevity) as the summation of friction losses in pipes and local losses due: (a) water entering and leaving the pipe network, (b) pipe bends, and (c) valves in the apparatus (based on formulae provided by White, 2016); and (3) unavoidable rise in the water level above the sand layer with higher Q, as required to discharge water through the overflow port(s). In combination, these factors led to nonlinear relationships between i and the driving head. For example, when the initial sand boil occurred (at the end of Stage 1), a sudden increase in Q (reaching 1.53 m^3/d) was observed (Fig. 5a) despite a small increase in i. This can be attributed to the high flux through the preferential pathway created by the sand boil, amounting to a reduction in the resistance to flow through the sand (and thus a sudden increase in Q by a small rise of i). The hydraulic behaviour of the system was more predictable in Stage 1. For example, Fig. 5a shows a linear increase in Q with i during Stage 1, prior to the sand boil formation. This is consistent with Darcy's Law. In Stage 2, by reducing i, Q decreased and reached 0.81 m^3/d. During Stage 3, Q was measured as 0.97 m^3/d when the sand boil was reformed. This was less than the measured Q (= 1.53 m^3/d) for the initial sand boil at the end of Stage 1 because the corresponding i for the sand boil reformation was lower than the required i for the inial sand boil.

Similar trends to those observed in the experimental results given in Fig. 5 were also observed in the previous laboratory experiments of Yang and Wang (2017), where q (= Q/A; referred to as the discharge velocity [LT^{-1}] in their paper) linearly increased with i before sand boil formation, while a sudden increase in q was measured when the sand boil formed. In Stage 2, Q values linearly decreased in time as the driving head was lowered, although only a small gradient drop was required prior to cessation of the boil. During Stage 3, Q increased continuously with time until the end of the experiment, while i values varied between 0.68 and 0.76, due in part to the widening in the sand boil diameter, as mentioned above. If the flow in the sand boil is treated as flow in a vertical pipe, it is possible to assess this relationship according to the Darcy-Weisbach formula (Bear, 1972). That is, for a constant i, as the pipe diameter (here, the diameter of the sand boil) increases, the Darcy-Weisbach equation shows that the discharge increases non-linearly with the pipe's cross-sectional area. The increase in the pipe's diameter also causes lower wall friction effects (according to the Colebrook equation; White, 2016), causing a higher flow velocity in the pipe (sand boil), and, consequently, larger values of discharge. The relationship between the pipe's diameter, discharge and friction factor for a constant i (= 1) was calculated based on the Darcy-Weisbach and Colebrook equations (using $\mu = 10^{-3}$ Pa.s, $\rho_w = 996$ kg/m^3, and pipe wall roughness (ε [−]) assumed to be equal to $d_m = 0.18$ mm (Nikuradse (1933)). The assumption of $i = 1$ was adopted here for illustrative purposes. The resulting relationship between the pipe's diameter, discharge and friction factor is shown in Fig. S1 in the Supplementary material.

Fig. 6 presents the results of Experiment B ($L = 180$ mm). Fig. 6a shows the time series of i and Q. The water turned partly cloudy at $i = 0.71$ during Stage 1, and at the same moment, the first movement of sand particles was observed in the form of very small craters on the sand surface. This is in contrast to Experiment A, in which cloudiness was not observed during Stage 1. The initial sand boil was formed when i reached 1.15, and the water became very cloudy after the initial sand boil formation, signifying the rapid expulsion of fine material into the water column. This is apparent in Fig. 6b. A 2–4 mm heave (i.e., expansion in the sand layer thickness) was observed just before the initial sand boil occurred at the end of Stage 1. The sand boil initiated on the left side of the sand column and then moved towards the middle, which differs from Experiment A, in which the sand boil appeared in the center of the sand column cross-section.

During Stage 2, i remained almost constant (≈ 0.79) before the cession of the sand boil. The sand boil disappeared at the end of Stage 2 when i was lowered to 0.77, and reformed during Stage 3 when i was increased to 0.90. After the sand boil reformation, i remained in the range of 0.74–0.81 to maintain the sand boil, even though the driving head was continuously increased with time until the end of the experiment. This is similar to the nonlinear hydraulic behaviour observed in Experiment A.

As seen in Fig. 6a, Q linearly increased with i in Stage 1, while a sudden jump in Q (reaching 2.61 m^3/d) was observed when the initial sand boil formed at the end of this stage. Q was 1.41 m^3/d when the sand boil stopped at the end of Stage 2. The increasing driving head in Stage 3 caused Q to rise monotonically with time; from 1.36 to 8.22 m^3/d. Corresponding changes in i were small (varying between 0.74 and 0.90), similar to Experiment A. This is again attributed to the increase in the sand boil diameter,

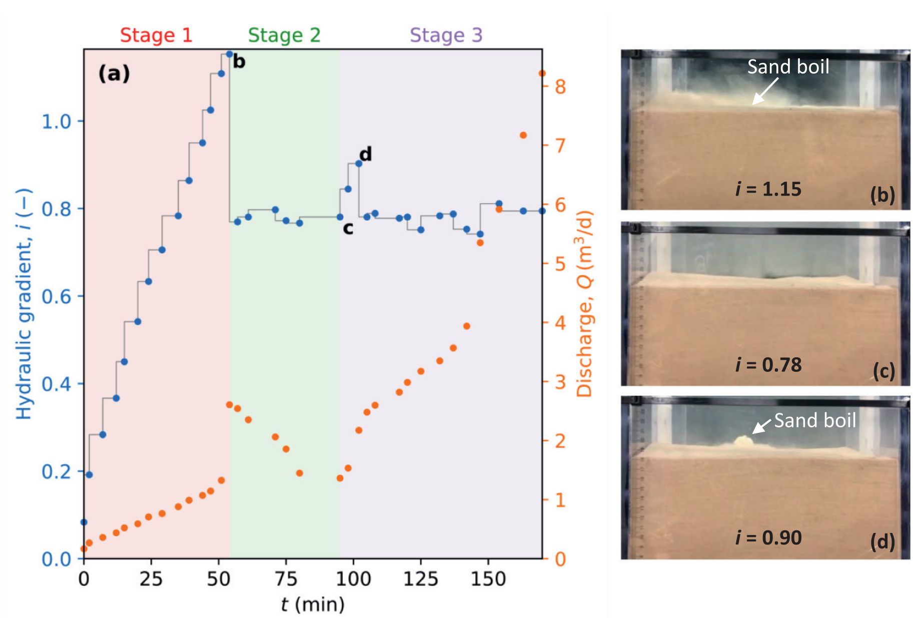

Fig. 6. Experiment B ($L = 180$ mm) results in terms of: (a) time series of i (blue circles) and corresponding Q (orange circles). Photographs of three key sand layer conditions are included at: (b) the first sand boil formation (the end of Stage 1), (c) sand boil cessation (the end of Stage 2), and (d) sand boil reformation during Stage 3. The light red, light green and light purple shaded areas in (a) represent Stages 1, 2 and 3 of the experiment, respectively.

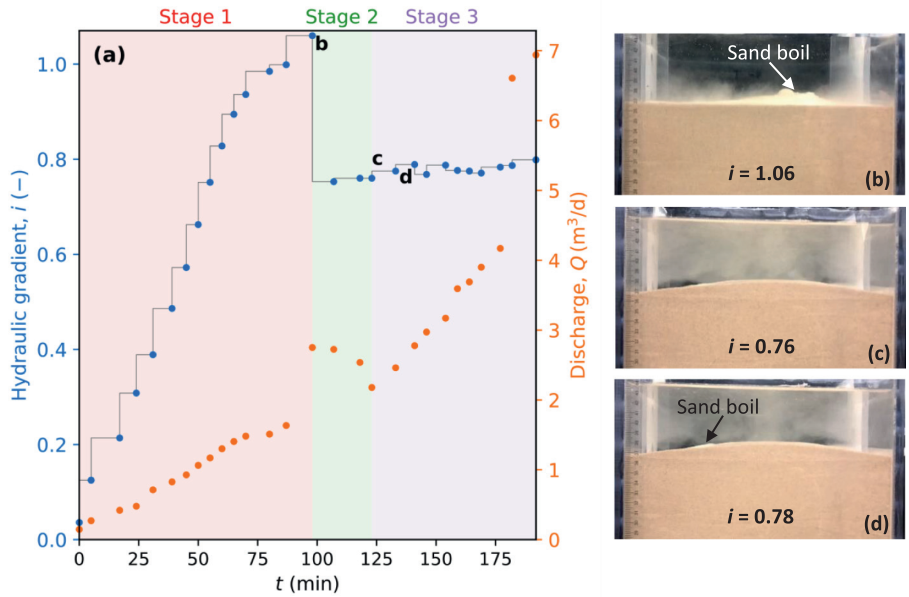

Fig. 7. Experiment C (L = 360 mm) results in terms of: (a) time series of i (blue circles) and corresponding Q (orange circles). Photographs of three key sand layer conditions at: (b) the first sand boil formation (the end of Stage 1), (c) sand boil cessation (the end of Stage 2), and (d) sand boil reformation during Stage 3. The light red, light green and light purple shaded areas in (a) represent Stages 1, 2 and 3 of the experiment, respectively.

reducing the resistance to discharge through the sand boil, and the other factors raised above for Experiment A. When the sand boil reformed during Stage 3, the measured Q (= 2.17 m^3/d) was lower than Q (= 2.61 m^3/d) for the initial sand boil at the end of Stage 1, similar to Experiment A. This is because a larger value of i was required for the initial sand boil to be occurred compared to the required i for the sand boil reformation.

Experiment C (sand thickness increased to 360 mm) produced qualitatively similar trends and patterns of behaviour as those observed in Experiments A and B. The time series of i and Q observed in Experiment C are shown in Fig. 7a. During Stage 1, the first sand particle movements at the sand surface were observed as very small craters when i reached 0.83, at which point the water column above the sand layer became cloudy. The initial sand boil formed when i reached 1.06, marking the end of Stage 1. A noticeable heave, in the form of an approximately 8 mm rise of the entire sand surface, occurred prior to the sand boil formation in Experiment C. The sand boil formed in the middle of the sand column cross-section, with fine particles expelled creating cloudiness in the overlying water body. The sand boil disappeared at the end of Stage 2 when i fell to 0.76. Once the sand boil stopped, the discharge through the sand column was clear, as can be seen by the layer of clear water immediately above the sand surface in Fig. 7c. The sand boil reformed at the beginning of Stage 3 when i was increased to 0.78. During Stage 3, i fluctuated between 0.78 and 0.80, showing highly nonlinear responses to the increasing driving head.

Fig. 7 shows that, similar to Experiments A and B, Q varied almost linearly with i in Stage 1, with a sudden rise in its value observed when the sand boil initially formed. By reducing i in Stage 2, Q decreased and reached 2.17 m^3/d at the end of Stage 2. During Stage 3, Q monotonically increased with time even though the fluctuation in i values was small (i.e., ranging from 0.77 to 0.80). As mentioned before, this can be because of widening the sand boil diameter (although it was not possible to measure the sand boil diameter during the experiments due to the cloudiness of the water when the sand boil occurred), increasing head losses in the apparatus, and raising the water level in the sand column during Stage 3 of the experiment. Similar to Experiments A and B, Q (= 2.46 m^3/d) for the reformed sand boil in Stage 3 was less than Q (= 2.68 m^3/d) for the initial sand boil in Stage 1 due to smaller i for the reformed sand boil with respect to i for the initial sand boil.

Table 2 summarizes measured values of i for Experiments A, B and C associated with the various events of each case, namely: (a) the first visible sand movement (i.e., suffusion), (b) formation of the initial sand boil, (c) sand boil cessation, and (d) sand boil reformation. Theoretical values of i_{cr} obtained from Terzaghi (1922), Knorre (1925), and Zamarin (1931)

Table 2
Experimental values of i associated with sand boil events in Experiments A, B and C. Theoretical values of i_{cr} based on Terzaghi (1922), Knorre (1925) and Zamarin equations are also given.

	i (−)		
	Experiment A (L = 90 mm)	Experiment B (L = 180 mm)	Experiment C (L = 360 mm)
First visible sand movement (suffusion)[a]	0.76	0.71	0.83
Initial sand boil formation[b]	0.97	1.15	1.06
Sand boil cessation[c]	0.65	0.78	0.76
Sand boil reformation[d]	0.76	0.90	0.78
Terzaghi's (1922) equation[e]	1.02	1.02	1.02
Knorre's (1925) equation[e]	1.65	1.65	1.65
Zamarin's (1931) equation[e]	1.21	1.21	1.21

[a] Based on results in Tables S1, S2 and S3 (the Supplementary Material) for Experiments A, B and C, respectively.
[b] Corresponding to point “b” in Figs. 5a, 6a and 7a for Experiments A, B and C, respectively.
[c] Corresponding to point “c” in Figs. 5a, 6a and 7a for Experiments A, B and C, respectively.
[d] Corresponding to point “c” in Figs. 5a, 6a and 7a for Experiments A, B and C, respectively.
[e] i_{cr} calculated by equations provided in Table 1 using n = 0.38 and G_s = 2.65.

equations using $n = 0.38$ and $G_s = 2.65$. However, only Terzaghi's (1943) i_{cr} (suggested by Yang and Wang (2017) and Petrula et al. (2019) as the value of i_{cr} for internally stable soils) were compared with experimental data. Table 2 shows that the required i for the first sand particle movement (i.e., suffusion) in all experiments is overestimated by Terzaghi's (1922) theory (for i_{cr}) by 35%, 44%, and 23% in Experiments A, B and C, respectively. Terzaghi's (1922) equation for i_{cr} overestimates (by 5 %) the required i for the initial sand boil formation in Experiment A (in which the sand layer is thinnest). However, for thicker sand layers, Terzaghi's (1922) equation underestimates the corresponding i values (by 11% and 4% for Experiments B and C, respectively) for initial sand boil formation. In all three experiments, the measured values of i required for the sand boil reformation were smaller (than that required for the initial sand boil to form) by 22%, 22% and 26% in Experiments A, B and C, respectively. Terzaghi's (1922) equation for i_{cr} overestimates i values for sand boil reformation by 34%, 13% and 31% for Experiments A, B and C, respectively. Note that Table 2 does not show a direct relationship between sand layer thickness and measure i values for the initial sand boil formation, sand boil cessation, and sand boil reformation. For example, Experiment B, with the medium sand layer thickness (180 mm), required larger i values for the initial sand boil formation, sand boil cessation, and sand boil reformation compared to Experiments A (with the thinnest sand layer; 90 mm) and C (with the thickest sand layer; 360 mm). However, the results clearly show the importance of the layer thickness and the history of sand boil formation in calculating i_{cr} (i.e., critical hydraulic gradient) required for sand boil formation.

4. Conclusions

Sand column experiments were conducted under controlled laboratory conditions to recreate sand boils in internally stable sand of varying layer thickness. The experimental results showed that thicker sand layers required higher hydraulic gradients (i) for sand boil formation than those predicted by Terzaghi's (1922) equation for the critical hydraulic gradient for fluidization (i_{cr}). This reveals the importance of sand layer thickness in sand boil formation, which is neglected in existing formulae for predicting the conditions under which a sand boil may arise.

Experimental observations prior to the sand boil emergence indicate that heave is more likely to be a pre-curser to sand boil formation in experiments with a thicker sand layer (360 mm), with no heave observed for the thinnest sand layer (90 mm).

Measured i values required for sand boil reformation, after a period of reduced i causing boil cessation, were lower than those needed to create the first occurrence of a sand boil. This indicates the importance of the sand boil history in their occurrence and cessation, adding to the known behaviour of sand boils – at least for those in which a relatively uniform body of sand is considered, similar to beach boils (i.e., rather than situations in Dutch polders where sand is overlain by cohesive sediments, such as peat). We hypothesize that this hysteretic behaviour in sand boil occurrence is due to changes in sand compaction after the formation of the first sand boil, although further testing is needed to quantify this effect. The current study examined the predictability of the famous Terzaghi's (1922) theory of sand layers of varying thickness, as well as under conditions where the driving head varies in time. The results highlight the need for further investigation to improve the existing theory to account for these effects. The findings of this study provide a better understanding of sand boil formation mechanisms of particular relevance to those found in intertidal zones and downstream of water-retaining structures, where head conditions are likely to vary over time.

CRediT authorship contribution statement

Amir Jazayeri: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software. **Adrian D. Werner:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Software, Methodology. **Thomas Jenkins:** Data curation, Investigation, Methodology, Project administration.

Data availability

I have added the laboratory data in the Supplementary Material.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary Material

Supplementary Material to this article can be found online at https://doi.org/10.1016/j.scitotenv.2023.163235.

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